THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL CHAPTER TEST SET 1

Chapter 4: **Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The atomic number of carbon is: 1

(a) 6 **(b)** 8 **(c)** 4 **(d)** 12

2. A single bond between two carbon atoms is found in: 1

(a) benzene **(b)** ethane **(c)** ethene **(d)** ethyne

3. Diamond and graphite are: 1

(a) alloys **(b)** compounds

(c) isomers **(d)** allotropes of carbon

4. The valency of carbon is: 1

(a) 4 **(b)** 3 **(c)** 6 **(d)** 2

5. A double bond is present in: 1

(a) methane **(b)** propane **(c)** ethene **(d)** ethane

6. Which allotrope of carbon conducts electricity? 1

(a) fullerene **(b)** graphite **(c)** diamond **(d)** none

7. The simplest hydrocarbon is: 1

(a) ethene **(b)** ethyne **(c)** ethane **(d)** methane

8. A triple bond is present in: 1

(a) ethyne **(b)** ethane **(c)** ethene **(d)** methane

9. Which is the hardest substance? 1

(a) fullerene **(b)** diamond **(c)** coal **(d)** graphite

10. The formula of methane is: 1

(a) C_2H_4 **(b)** CH_4 **(c)** C_2H_6 **(d)** C_3H_8

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is the valency of carbon? Why? 2

12. Define isomers. 2

13. Write the general formula of alkane and alkene. 2

14. Define a hydrocarbon. Give two examples. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the molecular formulas of the first three alkanes with names. 3

16. State three properties of carbon compounds. 3

17. Name the products of ethanoic acid and sodium hydroxide reaction and write the equation. 3

OR

Write the names of the first three alkenes with formulas. 3

18. State three uses of ethanol. 3

OR

What are soaps and detergents? Give one difference. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Draw electron dot structures of methane, ethene and water. State number of bonds. 5

OR

Describe reactions of ethanoic acid with sodium hydroxide, sodium carbonate and ethanol. 5

20. Differentiate between saturated and unsaturated hydrocarbons (five points). 5

OR

Explain how soap cleans cloth. Draw a micelle in words. What is the problem with hard water? 5

END OF QUESTION PAPER — Total: 40 Marks

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